An Improved Magnetic Equivalent Circuit Model for Iron-Core Linear Permanent-Magnet Synchronous Motors

B. Sheikh-Ghalavand, S. Vaez-Zadeh, and A. Hassanpour Isfahani

School of Electrical and Computer Engineering, Faculty of Engineering, University of Tehran, Tehran, Iran

The aim of this work is to establish an accurate yet simple method for predicting flux density distribution and iron losses in linear permanent-magnet synchronous motors (LPMSMs) for iterative design procedures. For this purpose, an improved magnetic equivalent circuit for calculation of the teeth and yoke flux densities in the LPMSMs is presented. The magnetic saturation of iron core is considered by nonlinear elements and an iterative procedure is used to update these elements. The armature reaction is also taken into account in the modeling by flux sources located on the teeth of motors. These sources are time dependent and can model every winding configuration. The relative motion between the motor primary and secondary is considered by wisely designing air gap elements simplifying the permeance network construction and preventing permeance matrix distortion during primary motion. Flux densities in different load conditions are calculated by means of the proposed model. The effects of saturation and armature reaction on the flux density distribution are shown in detail. Using these flux densities, iron losses in the motor are examined and its variations versus motor parameters are then studied. All results obtained by proposed model are verified by finite-element method based on an extensive analysis.

***Index Terms*—Armature reaction, finite-element method, iron loss, linear permanent-magnet synchronous motor, magnetic equivalent circuit, saturation.**

I. Introduction

LINEAR permanent-magnet synchronous motors (LPMSMs) are widely used in industrial applications due to their superior performance [1]. Iron loss evaluation of LPMSMs is required for analysis, design, and optimization of these motors. On the other hand, accurate modeling of flux density distribution in different parts of a machine, especially in teeth, is critical for iron loss calculation in LPMSMs. Different methods are employed to carry out the calculation including analytical methods, numerical methods, and magnetic equivalent circuit (MEC) based methods.

Analytical methods develop and solve Maxwell equations in different parts of a machine [2]–[9]. These methods try to find out closed form expressions for magnetic fields and losses in a motor. However, they cannot consider iron saturation which has significant effect on magnetic fields and losses of motors. They can hardly model the slot effect and magnetic flux leakage by some sort of approximation.

Numerical methods such as the finite-element method (FEM) are widely used for field and loss evaluation in permanent-magnet (PM) machines [10]–[14]. Although they are very accurate, they are time consuming and can hardly be used in iterative motor design optimizations and initial design procedures where several parameters change in wide ranges.

MEC has been known as an effective means for modeling flux density distribution of permanent-magnet (PM) machines for many years. The MEC based methods can model most phenomena related to PM machines such as nonlinearity, relative motion, armature reaction, eddy currents, leakage flux, etc. [15].

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MEC methods have been improved steadily in recent years, and now feature many capabilities. Simple MECs are sometimes used in designing PM motors by modeling only magnet pole and air gap reluctances which are inaccurate [16]–[19]. The capability of modeling leakage fluxes has been included to some MECs by different means, resulting in better estimations of flux densities [20]–[23]. However, these models do not consider magnetic saturation in motor teeth. The nonlinearity of iron properties has been included in some MECs, resulting in relatively accurate flux density predictions in iron parts of machines [24]–[27].

In the mentioned MECs, the effect of motor primary slots on the air gap flux density distribution is mostly overlooked. This effect can be considered in an MEC by a very large number of elements. Such an MEC is represented by almost 120 types of equivalent circuit elements [28]. A 3-D magnetic circuit is also proposed for analysis of LPMS motors which considers leakage fluxes and presents flux density distribution in detail [29], [30]. The calculation of force and torque has also been carried out based on the 3-D MEC [31]. Several kinds of hybrid MECs are developed and extended in corporation with FEM resulting in very accurate filed and force predictions [32]–[35]. However, these models are too complicated to be used in an iterative motor design procedure with different stages that need many variations in motor structure and dimensions to reach a final design.

A relatively simple and accurate magnetic equivalent circuit is proposed for PMLS motors together with a piecewise linear interpolation scheme to model air gap flux density distribution produced by PM poles by a mathematical function [36]. This method considers saturation as well as end teeth effects and leakage fluxes. However, the method suffers from two problems. It does not include armature reaction and relative motion of primary.

Although some researchers have presented methods to consider motion in PM motors, the change of reluctance network during motion adds to the complexity of solving equations [37], [38], [15].

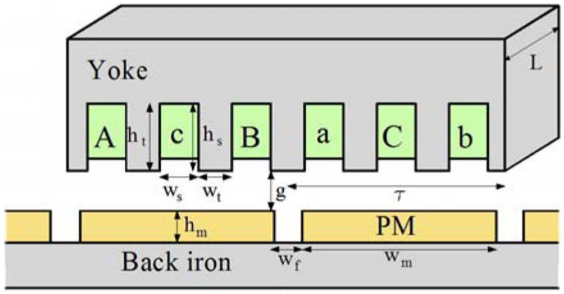

Fig. 1. Schematic view of a linear permanent-magnet synchronous machine.

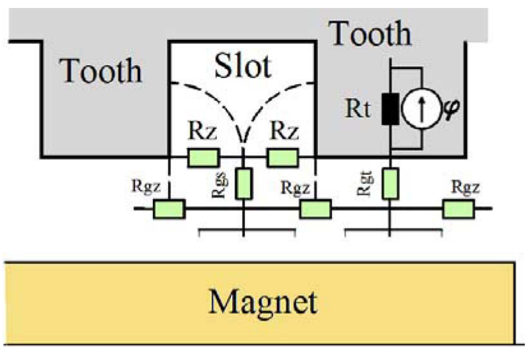

Fig. 2. Magnetic equivalent circuit for a part of machine primary.

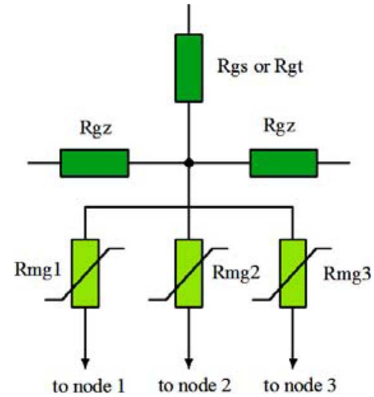

Fig. 3. Air gap model of MEC.

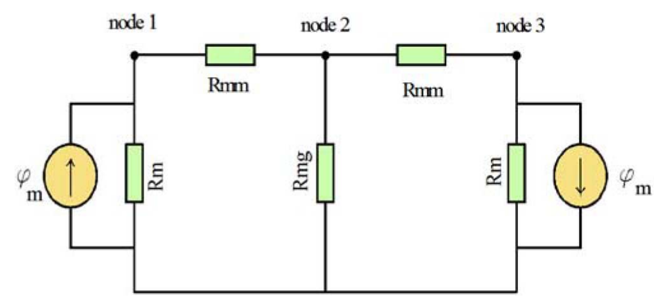

Fig. 4. Magnetic equivalent circuit for machine secondary.

S. Han *et al.* compare some MECs showing the lack of a comprehensive and simple model [15].

This paper presents a MEC of LPMSMs including saturation and slots effect as well as distributed windings, armature reaction, and primary motion, aimed at iron loss calculation. The structure of proposed MEC is fixed during the primary motion resulting in a fixed reluctance network matrix. This feature leads to a simple and time efficient solution of the MEC regarding the motion. The proposed model also provides a flux density distribution function in machine teeth. The model is employed to calculate iron losses in a LPMSM. Variation of iron losses with motor parameters are also studied using the proposed model. Finally, the method is verified by FEM and a good agreement between the results obtained by the two methods is observed.

II. Magnetic Equivalent Circuit of the LPMSM

In this section an improved MEC is proposed for LPMS motors. It is intended to retain much of the salient features of extensive and complex models despite the relative simplicity of the proposed MEC.

A two-pole LPMS machine is shown in Fig. 1. The proposed MEC for this motor consists of three main parts, i.e., primary, air gap, and secondary. A part of primary is shown in Fig. 2. It is seen that a tooth is modeled by a nonlinear reluctance indicated by black color, considering the saturation and a flux source for modeling the armature reaction.

The air gap is divided into two parts. The first part is between the primary and the middle of the air gap which is attached to primary by perpendicular and tangential constant valued air gap reluctances. Leakage reluctances also exist in the air gap network which model the longitudinal path of flux in the air gap.

The second part is between the secondary and the middle of the air gap. The nodes under the teeth and in the middle of the air gap are connected to three nodes of secondary with variable reluctances changing with regard to the relative position of primary to secondary. Therefore, during motion only the values of this part of MEC will change and the structure of reluctance network and its matrices dimensions are kept fixed. The air gap model is shown in Fig. 3. One pole pair of the secondary magnetic circuit is shown in Fig. 4. It consists of a magnets model and the flux leakage between two adjacent magnets.

The values of these reluctances can be calculated by applying Ampere's law. The reluctance values of primary network are given by

$$R_t = \frac{h_t}{\mu_0 \mu_{ri}(B_t) w_t L} \tag{1}$$

$$R_z = \frac{w_s}{\mu_0 h_s L} \tag{2}$$

where w_s and w_t denote slot width and tooth width, h_s and h_t represent slot depth and tooth height, and L is the motor width respectively as shown in Fig. 1.

Also, $\mu_{ri}(B_t)$ is the relative permeability of iron as a function of flux density to consider the saturation. The B-H curve of iron used in this paper is shown in Fig. 5.

The flux source of armature reaction varies in each tooth. For the winding arrangement of Fig. 1, the current sources are

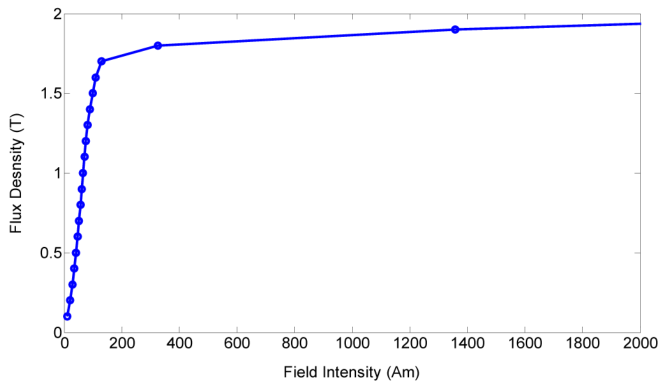

Fig. 5. B-H curve of primary and secondary iron.

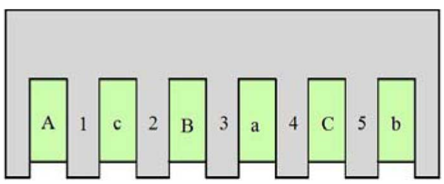

Fig. 6. Primary with numbered teeth.

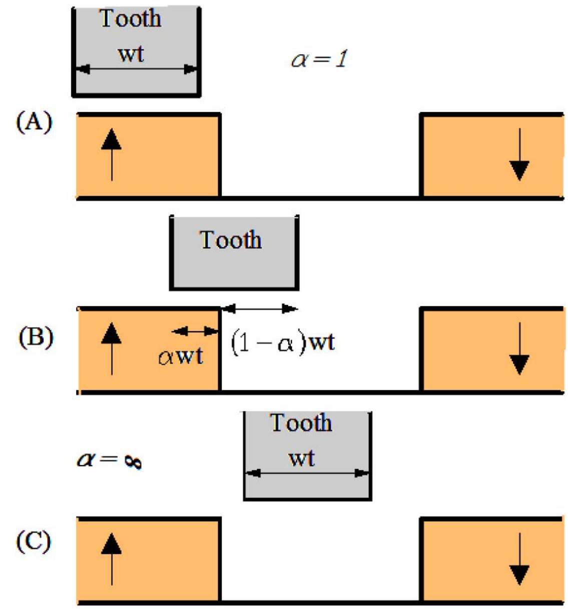

Fig. 7. Relative position of tooth with respect to PMs locations.

located on the middle teeth and calculated as follows:

$$\begin{aligned}\varphi_1 &= -\frac{1}{3}n_c i_A R_{t1}^{-1}\\ \varphi_2 &= \frac{1}{3}n_c(-i_A + i_C)R_{t2}^{-1}\\ \varphi_3 &= \frac{1}{3}n_c(-i_A + i_C - i_B)R_{t3}^{-1}\\ \varphi_4 &= \frac{1}{3}n_c(-i_B + i_C)R_{t4}^{-1}\\ \varphi_5 &= -\frac{1}{3}n_c i_B R_{t5}^{-1}\end{aligned} \tag{3}$$

where φ_i and R_{ti} are flux source and reluctance of ith tooth respectively, n_c is the number of coil turn in each slot, and i_{ABC} represents the current of phase A, B, and C as follows:

$$\begin{aligned}i_A &= i_m \sin\left(\frac{x\,\pi}{\tau} + \phi_0\right)\\ i_B &= i_m \sin\left(\frac{x\,\pi}{\tau} - \frac{2\pi}{3} + \phi_0\right)\\ i_C &= i_m \sin\left(\frac{x\,\pi}{\tau} + \frac{2\pi}{3} + \phi_0\right)\end{aligned} \tag{4}$$

where x, i_m, and ϕ_0 stand for primary position, maximum phase current, and an arbitrary phase angle relating to supply phase angle. The number of teeth is indicated in Fig. 6.

Air gap reluctances are also determined using Ampere's law as follows:

$$R_{\mathrm{gz}} = \frac{\tau_s}{2\mu_0 L\, g} \tag{5}$$

$$R_{\mathrm{gs}} = \frac{g}{2\mu_0 w_s L} \tag{6}$$

$$R_{\mathrm{gt}} = \frac{g}{2\mu_0 w_t L} \tag{7}$$

$$R_{\mathrm{gmi}} = \frac{g}{2\mu_0 \tau_s L\, \alpha_i} \tag{8}$$

where g and τ_s are air gap length and slot pitch, respectively, and α_i is a coefficient determined by relative position of each tooth and secondary PMs as illustrated in Fig. 7. Indeed, it is the ratio of the tooth portion located above the magnet to the tooth width and is infinite when none of the tooth is located above the magnet.

For the MEC of secondary, reluctances and flux source are determined as follows [15]:

$$R_{\mathrm{mm}} = \left[\frac{\mu_0 L}{\pi}\ln\left(1 + \frac{\pi\, g}{w_f}\right)\right]^{-1} \tag{9}$$

$$R_{\mathrm{m}} = \frac{h_m}{\mu_0 \mu_{rm} w_m L} \tag{10}$$

$$\varphi_m = \frac{H_C h_m}{R_{\mathrm{m}}} \tag{11}$$

where g, w_f, and h_m denote the physical air gap, the distance between two adjacent PM poles, and the PM height, respectively as shown in Fig. 1. Also, H_C and μ_{rm} are coercive force and relative permeability of PMs, respectively. The complete MEC is finally depicted in Fig. 8.

III. Numerical Solution of MEC

The MEC obtained in the previous section is relatively simple due to having a limited number of elements. Also, it is rather accurate due to the modeling of teeth, flux leakages, and saturation. Air gap flux density values under teeth and slots and the flux density in the middle of teeth are obtained by solving the MEC node equations:

$$[\Im] = [P]^{-1}[\phi] \tag{12}$$

where $[P]$, $[\phi]$, and $[\Im]$ are permeance matrix, flux source vector, and magnetomotive force (MMF) vector, respectively.

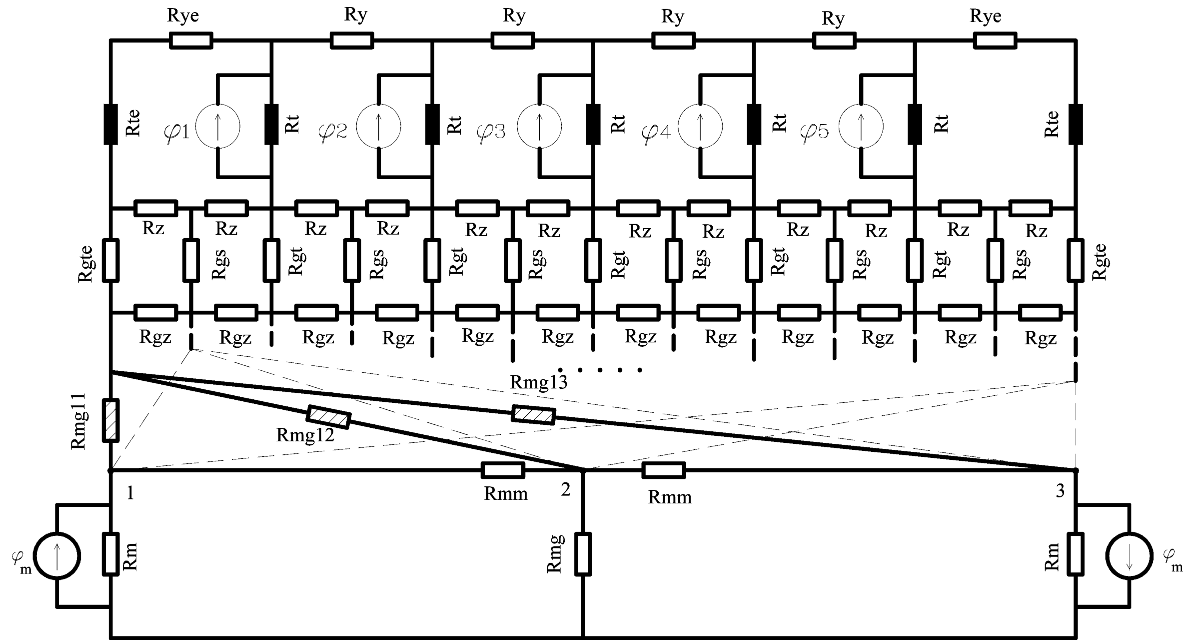

Fig. 8. Complete MEC model of linear permanent magnet synchronous machine.

There are only two flux sources in the MEC, so the flux source vector is formed as

$$[\phi] = [\varphi_m \quad 0 \quad -\varphi_m \quad 0 \quad \ldots \quad \varphi_1 \quad \ldots \quad \varphi_5 \quad \ldots \quad 0]^T. \tag{13}$$

Therefore, MMF of each node can be expressed as

$$F_i = (\hat{P}_{i1} - \hat{P}_{i2})\phi_m \tag{14}$$

whereF_i is a MMF of the ith node and $\hat{P}_{ij}$ is an element of the ith row and the jth column of $[P]$. The flux density in the branch is then given by

$$B_{ij} = \frac{(F_i - F_j)P_{ij}}{A_{ij}} \tag{15}$$

where B_{ij} and A_{ij} are the flux density and cross section area of the branch between the ith and jth nodes and P_{ij} is an element of the ith row and the jth column of permeance matrix.

Flux density distribution function (FDDF) for the middle of teeth is given by

$$B_y = \begin{cases} B_{\text{ti}} & \text{Tooth} \\ 0 & \text{Slots} \end{cases}. \tag{16}$$

The algorithm for nonlinear analysis of the motor is shown in Fig. 9. First, the dimensions and material properties of motor are initialized. Variables and air gap permeances and flux sources are calculated. Then, using a constant permeability for iron the permeance matrix of system is developed and solved. Flux density values in different parts of the motor are then computed and permeabilities of saturated parts are updated using B-H curve of iron parts and flux densities are recomputed for these regions.

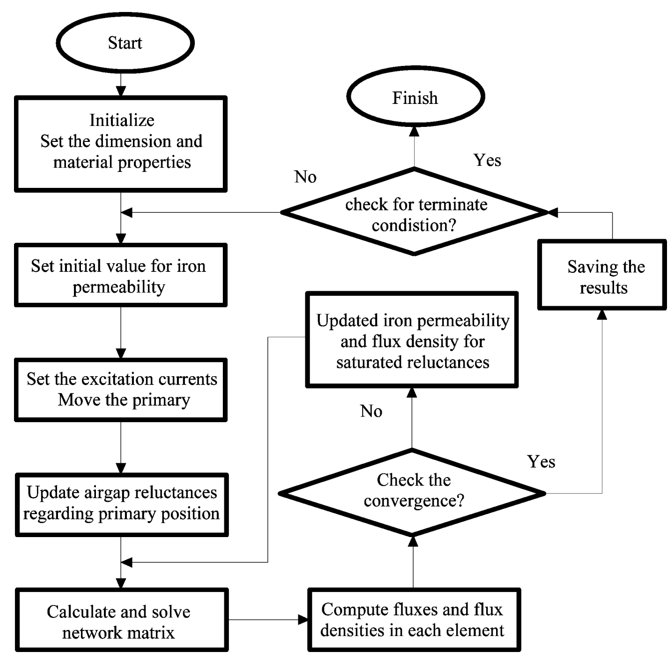

Fig. 9. The algorithm of nonlinear analysis.

The new permeability is calculated as follows:

$$\begin{aligned} \mu_{ri}(k) &= \hat{\mu}_{ri}^{0.1}(k) \cdot \hat{\mu}_{ri}^{0.9}(k) \\ \hat{\mu}_{ri}(k) &= \frac{B_i(k-1)}{\mu_0 H_i(k-1)} \end{aligned} \tag{17}$$

The convergence of flux density is then provided through an iterative process. The calculations may be repeated for other positions of the primary. Therefore, the flux density values are calculated in different parts of the system regarding the primary motion without any changes in the network matrix dimensions.

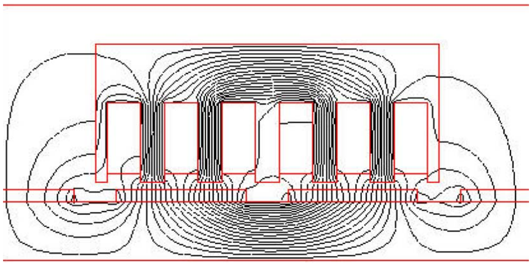

Fig. 10. Flux lines in LPMSM.

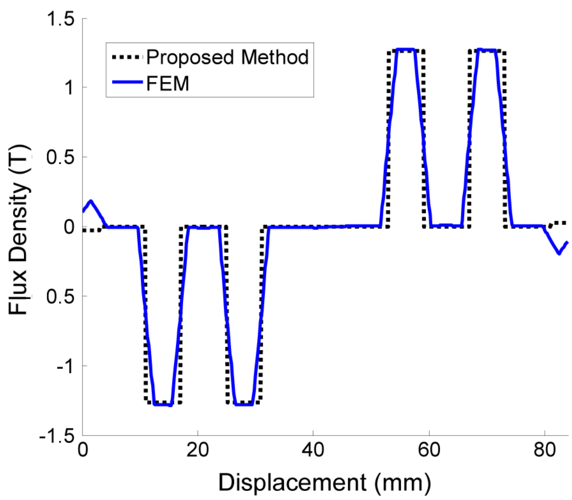

Fig. 11. Flux density in the middle of teeth (no load).

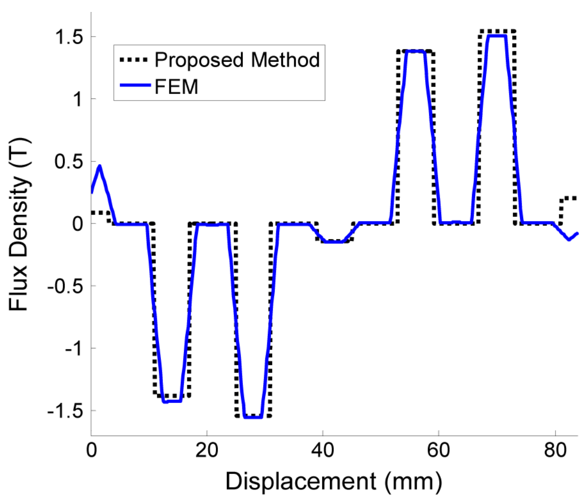

Fig. 12. Flux density in the middle of teeth (full load).

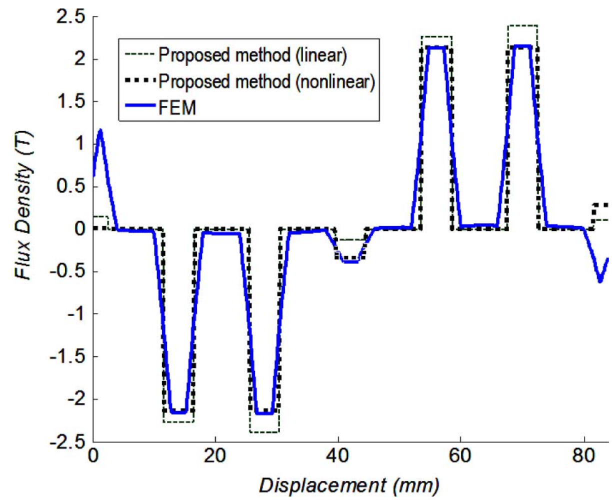

Fig. 13. Flux density in the middle of teeth (saturated motor).

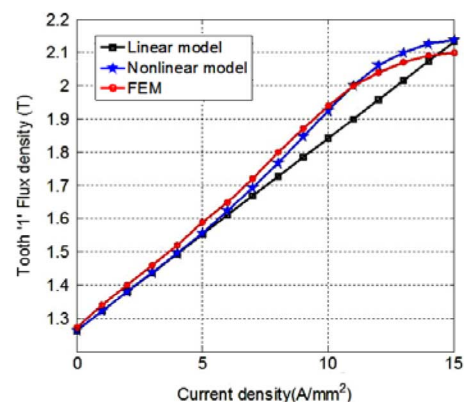

Fig. 14. Flux density variations of tooth '1' versus current density.

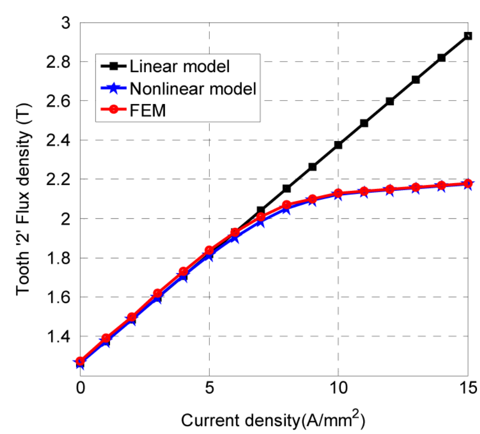

Fig. 15. Flux density variations of tooth '2' versus current density.

IV. Evaluation of Proposed MEC

A 2-D nonlinear time-stepping finite-element method is employed to verify the accuracy of the results obtained by the proposed method. Fig. 10 shows flux lines in the motor obtained by FEM. The flux density distribution in the middle of teeth in no-load conditions and full load conditions given by FEM and by the proposed method are compared in Figs. 11 and 12, respectively.

They show a good agreement between the results of FDDF and FEM. The saturation effect is then investigated in Figs. 13–16. Fig. 13 shows the flux density distribution in the middle of teeth when the motor is saturated. It is observed that the proposed nonlinear model is accurate enough while the linear model does not provide precise results.

The variation of flux density in different teeth of the motor in terms of current density is investigated in Figs. 14–16. Fig. 14 shows the flux density in tooth '1'. It is seen that in the low currents, linear model, and nonlinear model have the same accuracy. However, with an increase in the current, the flux density

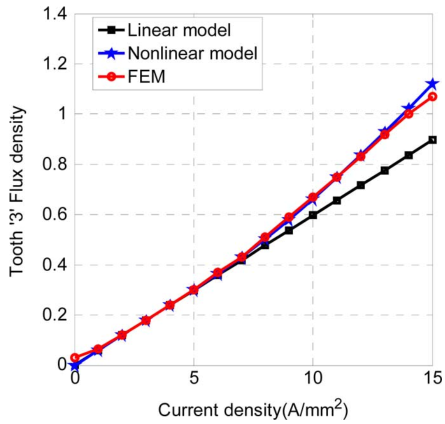

Fig. 16. Flux density variations of tooth '3' versus current density.

of nonlinear model becomes more than that of the linear model because tooth '2' experiences saturation and its reluctance increases so that the flux path moves to the other teeth like tooth of '1'. Finally, if the current increases further, the tooth '1' also goes to saturation and the rate of increases in the flux density of nonlinear model reduces.

Fig. 15 shows the flux density in the tooth '2' which is highly saturated by an armature current. It is seen that with the increase of current, the error between linear and nonlinear model also increases.

Fig. 16 shows the flux density in tooth '3' which does reach to the saturation level and therefore as described for tooth '1', the flux density of nonlinear model is more than that of the linear model. It is seen in all figures that the proposed nonlinear model enjoys acceptable accuracy in comparison with FEM results. Bearing in mind the desirable accuracy of the FEM results, it can be concluded that the proposed method is capable of modeling accurately the teeth flux density distribution along the machine.

V. Iron Loss Calculation

Having the flux density in different parts of the machine, the iron loss can be achieved. The iron loss is separated into two components, i.e., hysteresis losses and eddy-current losses. Therefore, we have

$$P_t = P_h + P_e = k_h B_m^{\beta} f + k_e B_m^2 f^2 \quad (W/m^3) \tag{18}$$

where $\beta = 1.8$–2.2 and k_h and k_e depend on the laminated material and B_m is maximum value of flux density. Remember that (18) is only true for sinusoidal flux density, which is not always the case. In this situation, while the hysteresis loss is easy to be found as it depends only on the peak value of flux density, the eddy-current losses cannot be estimated accurately [39]. It is demonstrated that the eddy-current losses calculated by (18) is much lower than a measured value [40]. Therefore, it is convenient to represent eddy-current losses as a function of flux density in time domain as follows:

$$P_e = \frac{2k_e}{T} \int_0^T \left(\frac{dB(t)}{dt}\right)^2 dt \quad (\mathrm{W/m^3}) \tag{19}$$

where T is a period of flux density curvature.

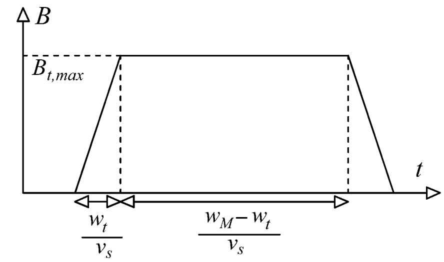

Fig. 17. Flux density in the teeth.

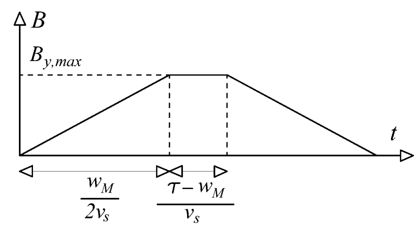

Fig. 18. Flux density in yoke.

The flux density in the teeth and yoke of an LPMSM is approximately like as the ones shown by Figs. 17 and 18. The maximum value of flux density is determined by the proposed MEC. Therefore, eddy-current loss density for teeth and yoke are given by

$$P_{et} = 12qk_e\left(\frac{v_s}{\tau}B_{t,\max}\right)^2 \quad (\mathrm{W/m^3}) \tag{20}$$

$$P_{ey} = 8k_e\frac{\tau}{w_m}\left(\frac{v_s}{\tau}B_{y,\max}\right)^2 \quad (\mathrm{W/m^3}) \tag{21}$$

where v_s, τ, w_m, and q are linear speed, pole pitch, magnet width, and number of slots per pole per phase, respectively. The $B_{t,\max}$ and $B_{y,\max}$ are maximum value of flux density in the teeth and yoke shown in Figs. 17 and 18, respectively. The above equations do not consider the longitudinal component of eddy-current losses. For this reason, a coefficient which is about 1.1–1.2 is put into (20) and (21), [39]. Therefore, total eddy-current loss in motor will be as follows:

$$\begin{aligned} P &= 12qk_ek_c\left(\frac{v_s}{\tau}B_{t,\max}\right)^2 \\ &\quad \cdot V_{\mathrm{th}} + 8k_ek_c\frac{\tau}{w_m}\left(\frac{v_s}{\tau}B_{y,\max}\right)^2 \cdot V_y(W) \end{aligned} \tag{22}$$

$$\begin{aligned} P_e &= 12qk_ek_c\left(\frac{v_s}{\tau}B_{t,\max}\right) \\ &\quad \cdot V_{\mathrm{th}} + 8qk_ek_c\frac{\tau}{w_M}\left(\frac{v_s}{\tau}B_{y,\max}\right) \cdot V_y \quad (W). \end{aligned} \tag{23}$$

Iron losses are also calculated by nonlinear time-stepping FEM. Eddy-current and hysteresis losses are calculated using FE formulation [39].

The results obtained from MEC are compared with the results obtain from FEM in Table I for a motor whose specification is listed in the Appendix. It is seen that only a 3% error exists between the proposed MEC iron loss and that obtained by the nonlinear time-stepping FEM. It is also seen that the MEC method gives lower hysteresis losses because in MEC method the flux

TABLE I
MEC AND FEM COMPARISON

Method	MEC	FEM	Error
Hysteresis loss (W)	7.7	8.27	7%
Eddy Current loss (W)	2.37	2.13	-11%
Total loss (W)	10.1	10.4	3%

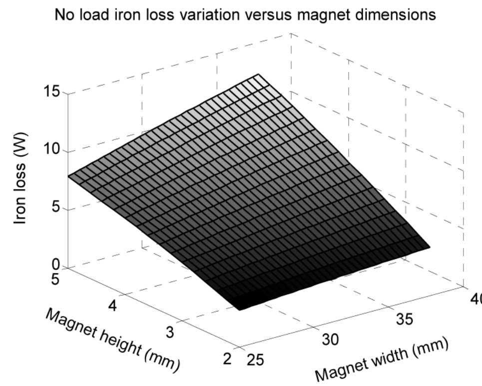

Fig. 19. Iron loss variations in terms of magnet dimensions at no-load conditions.

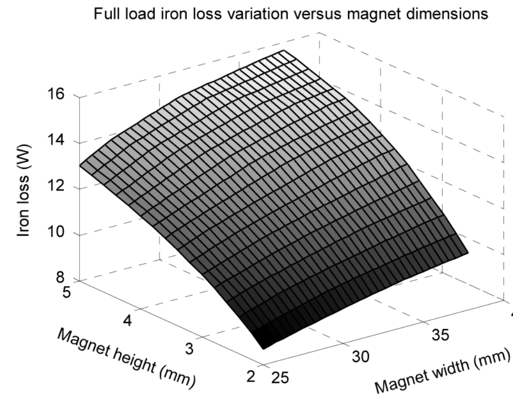

Fig. 20. Iron loss variations in terms of magnet dimensions at full-load conditions.

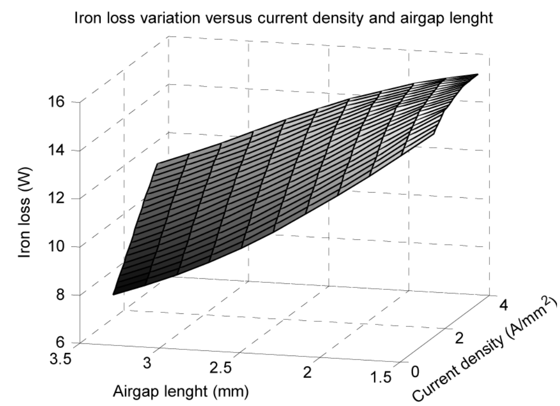

Fig. 21. Iron loss variations in terms of air gap length and primary current density.

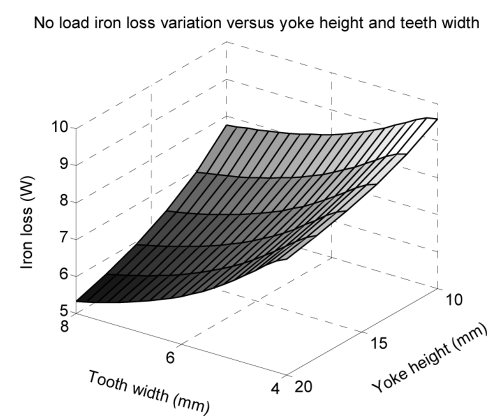

Fig. 22. Iron loss variations in terms of teeth width and yoke height at no-load conditions.

density distribution is assumed to be uniform in all over each element, but in practice this distribution may not be uniform and the local saturation in sharp sections is occurred increasing the hysteresis losses in FEM.

However, eddy-current losses mainly depend on the shape of flux density waveform. The proposed method consider the trapezoidal waveform whereas in the real case this waveform has smoother shape which reduces the value of its derivation leading to lower eddy-current losses in FEM.

VI. Iron Loss Analysis

In this section using the proposed method, the effects of different parameters variation of iron loss are investigated through several 2-D and 3-D plots.

Magnet width and height, air gap length, teeth width, and yoke height are chosen to consider their effects. The effect of magnet width and magnet height on the iron losses for no-load conditions and full-load conditions are shown in Figs. 19 and 20. It is seen that an increase in the magnet width and the magnet height results in an increase in the loss. Fig. 20 shows the saturation effect as the peak of surface becomes rather curvature.

The effects of variations in air gap length and primary current on iron loss is depicted in Fig. 21. Although this figure shows the positive effect of large air gap on the iron loss reduction, this may results in a substantial reduction in out put power. Therefore, in design optimization of permanent magnet, the air gap length should be selected wisely to prevent both a reduction in output power and an increase in iron loss.

Effects of teeth width and yoke height on iron loss are illustrated in Figs. 22 and 23. These figures show that the iron loss is more sensitive to teeth width

VII. Conclusion

An improved magnetic equivalent circuit was presented for permanent-magnet linear synchronous motors to calculate magnetic field density and iron loss of such machines. The model takes into account the magnetic saturation of iron core as well as the armature reaction and the relative motion. The proposed method of primary source modeling gives us the opportunity of considering every winding configuration.

A novel air gap element formation is used simplifying the permeance network construction and preventing permeance matrix distortion during primary motion. Different motor characteristics were calculated by means of the proposed model. The accuracy of the proposed method in modeling of the machines was

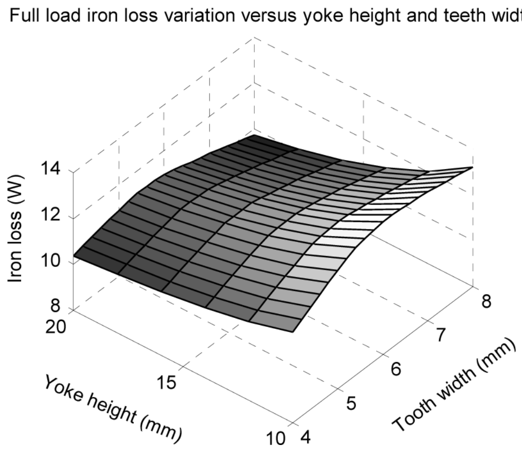

Fig. 23. Iron loss variations in the of teeth width and yoke height at full-load conditions.

TABLE II
MOTOR SPECIFICATIONS

Parameter	Value	Unit
Pole pitch	42	mm
Air gap length	2	mm
Magnet width	31.5	mm
Magnet height	4	mm
Tooth height	30	mm
Tooth width	6	mm
Slot pitch	14	mm
Motor width	84	mm
Magnet type	N33	-
Yoke height	10	mm
k_h	50	-
k_e	0.05	-

verified by finite-element method. The results show that the proposed method accurately models the saturation and armature reaction which are important in iron loss calculation. Finally, iron losses have been calculated by the proposed method and verified by FEM. The results show that the proposed model is very accurate for iron losses calculations in spite of its simplicity and time efficiency in solving procedure. Therefore, the proposed MEC is suitable for design and optimization of motors regarding iron losses.

APPENDIX

Specifications of the analyzed motor are as shown in Table II.

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Babak Sheikh-Ghalavand received the B.Sc. degree in electrical engineering from Ahvaz University, Ahvaz, Iran, in 2005 and the M.Sc. degree in electrical engineering from the University of Tehran, Tehran, Iran, in 2008.

He is currently working at Peiman Co., Tehran, Iran.

Sadegh Vaez-Zadeh (S'95–A'03–SM'05) received the B.Sc. degree from Iran University of Science and Technology, Tehran, Iran, in 1985 and the M.Sc. and Ph.D. degrees from Queen's University, Kingston, ON, Canada, in 1993 and 1997, respectively, all in electrical engineering.

He was with several research and educational institutions in different positions before joining the University of Tehran as an Assistant Professor in 1997, where he became an Associate Professor in 2001 and a Full Professor in 2005. He has served the university as the Head of Power Division from 1998 to 2000 and currently is the Director of Advanced Motion Systems Research Laboratory which he founded in 1998 and the Director of Electrical Engineering Laboratory since 1998. His research interests include advanced rotary and linear electric machines and drives, magnetic levitation, electric vehicles, and power system analysis and control. He has authored or co-authored over 150 research papers in these areas.

Prof. Vaez-Zadeh is a member of IEEE PES Motor Sub-Committee and Power System Stability Control Sub-Committee. He has received a number of awards domestically including a best paper award from Iran Ministry of Science, Research and Technology in 2001 and a best research award from the University of Tehran in 2004. He is an Editor of IEEE TRANSACTIONS ON ENERGY CONVERSION and an Editor of *Journal of Faculty of Engineering*, the oldest technical research journal in the Middle East. He is also a founding member of the editorial board of *Iranian Journal of Electrical and Computer Engineering*. He has been active in IEEE sponsored conferences as a member of technical and steering committees, session chair, etc.

Arash Hassanpour Isfahani was born in Isfahan, Iran, in 1980. He received the B.Sc. degree in electrical engineering from Isfahan University of Technology, Isfahan, Iran, in 2002 and the M.Sc. degree (with honors) in electrical power engineering from the University of Tehran, Tehran, Iran, in 2005, where he is currently working toward the Ph.D. degree. His research interests include design, modeling, and control of electrical machines, finite-element analysis of electromagnetic devices, and magnetic levitation.